ModelIt!

Teacher's Guide

Why Do Pandemics Spiral Out of Control?

Modeling Infectious Disease Transmission and Public Health Interventions

Quick Reference

Grade Level	Grade 11L1
NGSS Standard	HS-LS2-2, HS-LS4-6
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS2-2, HS-LS4-6

Use mathematical representations to support and revise explanations based on evidence about factors affecting biodiversity and populations in ecosystems of different scales. Create or revise a simulation to test a solution to mitigate adverse impacts of human activity on biodiversity.

Three-Dimensional Learning

Science Practice	**Using Mathematics and Computational Thinking** — Students use the SIR (Susceptible-Infected-Recovered) model framework and R0 calculations to predict epidemic dynamics, evaluate intervention timing, and compare outbreak scenarios mathematically.
Disciplinary Core Idea	**LS2.A Interdependent Relationships in Ecosystems / LS4.D Biodiversity and Humans** — Disease transmission follows mathematical patterns governed by pathogen-host interactions. Human activities (population density, travel, intervention policies) fundamentally alter these dynamics.
Crosscutting Concept	**Cause and Effect** — Students trace the causal chain from pathogen R0 through population susceptibility to epidemic outcome, identifying how interventions at specific points in the chain produce dramatically different results depending on timing.

CAST Testing Alignment

MULTIPLE CHOICE:

A new pathogen has R0 = 4.0. Using the herd immunity threshold formula (1 - 1/R0), what percentage of the population must be immune to prevent sustained transmission? A) 25%, B) 50%, C) 75%, D) 90%

CONSTRUCTED RESPONSE:

Using your model, explain why a 20-day delay in implementing social distancing can result in 5-10 times more total infections than early implementation. Connect your answer to exponential growth, the effective reproduction number, and the time-dependent shrinking of the susceptible population.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Pandemics have shaped human history more than wars, and they follow mathematical laws as precise as any in physics. The SIR (Susceptible-Infected-Recovered) model, developed in the 1920s, remains the foundation of epidemiological modeling because it captures the essential dynamics: a susceptible population that shrinks as people are infected, an infectious population that grows and then declines, and a recovered population that provides immunity. Understanding these dynamics isn't just academic — it's the science that determines whether millions live or die.

The SIR Model Framework

The SIR model divides a population into three compartments: Susceptible (S) — those who can be infected, Infected (I) — those who are currently infectious, and Recovered (R) — those who have cleared the infection and are immune. The flow between compartments follows differential equations: dS/dt = -beta*S*I, dI/dt = beta*S*I - gamma*I, dR/dt = gamma*I, where beta is the transmission rate and gamma is the recovery rate. R0 = beta/gamma represents the basic reproduction number. When S is large (early epidemic), Re is close to R0 and infections grow exponentially. As S shrinks (people recover), Re falls until Re < 1 and the epidemic declines.

Exponential Growth and the Doubling Time

In the early phase of an epidemic, when nearly everyone is susceptible, infections grow exponentially: I(t) = I_0 * e^(kt), where k depends on R0 and the generation time. For COVID-19 (R0~2.5, generation time ~5 days), the doubling time was approximately 3-4 days in uncontrolled settings. This means 10 cases become 20, then 40, 80, 160, 320, 640, 1280 — from 10 to over 1,000 in just three weeks. The deceptive quiet of the early phase (only 10-100 cases) is precisely why pandemics catch populations by surprise.

Herd Immunity: Natural vs. Vaccination

Herd immunity occurs when enough people are immune that the effective reproduction number drops below 1. The threshold is 1 - 1/R0. For R0=2.5: threshold is 60%. For R0=15 (measles): threshold is 93%. Reaching this threshold through natural infection means 60-93% of the population must be infected — with all the associated hospitalization and death. Vaccination achieves the same mathematical result without the illness. This is why vaccination is the single most powerful tool in pandemic response: it reduces the susceptible population directly.

Why Timing Is Everything

Mathematical modeling consistently shows that the same intervention implemented 2-3 weeks earlier can reduce total infections by 50-80%. The reason is exponential growth: during those 2-3 weeks, the infected population doubles multiple times. Early social distancing when there are 100 cases means controlling 100 infections; the same measures 3 weeks later means controlling 1,600 infections. This creates the 'paradox of prevention': successful early intervention makes the threat seem like it was never serious, leading critics to claim the response was an overreaction. In reality, the low case count IS the evidence that early intervention worked.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify R0 value, vaccination rate, susceptible population, infection rate, recovery rate, and effective R as the key components of the epidemic transmission system.

E **Establish Relationships**

Students determine that R0 and vaccination rate are external inputs (pathogen biology and policy decisions), while susceptible population, infection rate, recovery rate, and effective R are internal dynamics governed by the SIR model equations.

V **Visualize & Model**

Students build a computational SIR model showing how infections flow from susceptible to infected to recovered, with the effective reproduction number declining as susceptible population shrinks.

E **Evaluate Outcomes**

Students run uncontrolled, vaccinated, and early vs. late intervention scenarios to observe how exponential growth, herd immunity, and intervention timing shape epidemic curves.

R **Revise & Extend**

Students add superspreader events, hospital capacity, or behavioral compliance to explore how real-world heterogeneity and human behavior modify the idealized SIR predictions.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"In January 2020, there were 282 confirmed COVID cases worldwide. By January 2021, there were 100 million. How does a number that small become a number that large?"*

Watch For

Show the COVID case count graph from January 2020 to peak. Point out the deceptive quiet of the early weeks. Ask: At what point would YOU have started worrying?

Slide 2: Learning Objectives

3 min

Say: *"Today we're modeling the mathematics of contagion — the same equations that epidemiologists use to predict, and try to prevent, the next pandemic."*

Watch For

Have students read objectives. Pre-teach R0 with the rice-on-a-chessboard analogy: if each grain doubles, how many squares until the board overflows? (Answer: 64 squares, final square has 9.2 quintillion grains.)

Slide 3: Big Question

3 min

Say: *"SARS in 2003: 8,098 cases, contained. COVID in 2020: 700 million cases, global pandemic. Both respiratory viruses. Both from China. What was different?"*

Watch For

Quick-write: What factors do you think determine whether an outbreak stays small or goes global? Share out and list responses.

Slide 4: LEVER Framework

2 min

Say: *"We're building the same model the CDC and WHO use to predict outbreaks. It's called the SIR model, and it's surprisingly simple — just three boxes and two arrows."*

Watch For

Draw S → I → R on the board. Explain: people flow from Susceptible to Infected to Recovered. The speed of the first arrow is the whole game.

Slide 5: Activity 1: Components

8 min

Say: *"We can choose how contagious the pathogen is and how many people are vaccinated. Everything else — infection rate, how fast it spreads, when it peaks — follows from those choices."*

Watch For

Guide sorting: R0 Value and Vaccination Rate are external (biology and policy). Susceptible Population, Infection Rate, Recovery Rate, and Effective R are internal (model dynamics).

Slide 6: Activity 2: Connections

8 min

Say: *"Here's the key insight: R0 starts as a constant, but the EFFECTIVE R changes every day because every new infection removes one person from the susceptible pool. The epidemic is fighting against its own success."*

Watch For

Students map the feedback loop: higher infection → lower susceptible population → lower effective R → eventually Re < 1 → epidemic declines. This is the negative feedback loop that ends all epidemics.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's run three worlds: no intervention, 60% vaccination, and early vs. late social distancing. Watch the epidemic curves — the differences will shock you."*

Watch For

Students predict peak infection day and total infections for each scenario. Run simulations. The timing comparison (Day 10 vs. Day 30 intervention) should produce the most dramatic difference.

Slide 8: Discoveries

5 min

Say: *"Twenty days. That's all it takes. The same intervention, implemented 20 days earlier, can cut total infections by 80%. In a pandemic, procrastination kills."*

Watch For

Show the side-by-side epidemic curves for early vs. late intervention. Calculate: at a 4-day doubling time, 20 days = 5 doublings = 32x more infections. The math is the argument.

Slide 9: STEM Challenge

12 min

Say: *"A new virus just showed up. R0 is 3.0. Cases are doubling every 4 days. No vaccine for 6 months. You're in charge. Design the response plan that saves the most lives while spending the least money."*

Watch For

Groups design phased response plans with intervention triggers, resource allocation, and communication strategies. Must justify timing decisions with model evidence. Present and defend plans.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** R0 Value, Vaccination Rate
- **INTERNAL:** Susceptible Population, Infection Rate, Recovery Rate, Effective R (Re)

Relationships

- **R0 Value to Infection Rate:** POSITIVE (+) — Higher R0 means each infected person transmits to more others, directly increasing the rate of new infections. A pathogen with R0=3 produces 3x more new infections per generation than one with R0=1.
- **Vaccination Rate to Susceptible Population:** NEGATIVE (-) — Higher vaccination rates directly reduce the susceptible population — each vaccinated person is removed from the pool of potential hosts. At 60% vaccination, only 40% of the population remains susceptible.
- **Susceptible Population to Effective R:** POSITIVE (+) — Re = R0 * (fraction susceptible). As the susceptible population shrinks (through vaccination or natural infection), the effective reproduction number drops proportionally. When Re falls below 1, the epidemic declines.
- **Infection Rate to Susceptible Population:** NEGATIVE (-) — Each new infection removes one person from the susceptible pool. Higher infection rates accelerate the depletion of susceptible individuals, which eventually reduces Re below 1 — this is the negative feedback loop that ends all epidemics.

Key Discoveries

- Epidemics grow exponentially, not linearly — each infected person infects R0 others, who each infect R0 more, creating a doubling cascade that makes early stages deceptively quiet before infections explode
- The herd immunity threshold (1 - 1/R0) determines the minimum vaccination coverage needed to prevent outbreaks — for R0=2.5, this is 60%, for R0=15 (measles), it's 93%
- Intervention timing is the most critical variable — the same intervention implemented 20 days earlier can reduce total infections by 50-80% because exponential growth means every day of delay allows the infected population to double
- All outbreaks eventually end because the susceptible population shrinks — the question is whether they end through controlled vaccination (few deaths) or uncontrolled infection (mass death and overwhelmed hospitals)

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Pandemics grow at a constant rate"	Pandemics grow EXPONENTIALLY in the early phase, not linearly or at a constant rate. This means the rate of new infections increases every day — each generation of infections is larger than the last by a factor of R0. This is why an outbreak that seems manageable (100 cases) can become catastrophic (100,000 cases) in just a few weeks. Students often graph pandemic growth as a straight line because linear thinking is intuitive. The model shows the actual exponential curve, which curves sharply upward — a shape that is qualitatively different from linear growth.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Herd immunity means NOBODY gets sick"	Herd immunity doesn't prevent all infections — it prevents sustained CHAINS of transmission. When the susceptible population drops below the critical threshold (1-1/R0), each infected person transmits to fewer than one other person on average. Individual infections still occur, but they can't sustain an epidemic chain — they fizzle out. Think of it like a fire: herd immunity doesn't prevent sparks, it prevents the forest fire by removing enough fuel (susceptible people) that flames can't spread.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"You can wait to see how bad it gets before responding"	This is the most dangerous misconception because it leads to the worst outcomes. Exponential growth means that by the time the threat is 'obvious' (thousands of cases, hospitals filling), the epidemic has been doubling for weeks and contains millions of infections in the pipeline. Waiting for visible evidence of severity guarantees a late response. The model shows this brutally: the same intervention at Day 10 (100 cases) vs. Day 30 (3,200 cases) results in 5-10x more total infections. In a pandemic, the evidence that justifies action is the mathematical model, not the case count you can see.